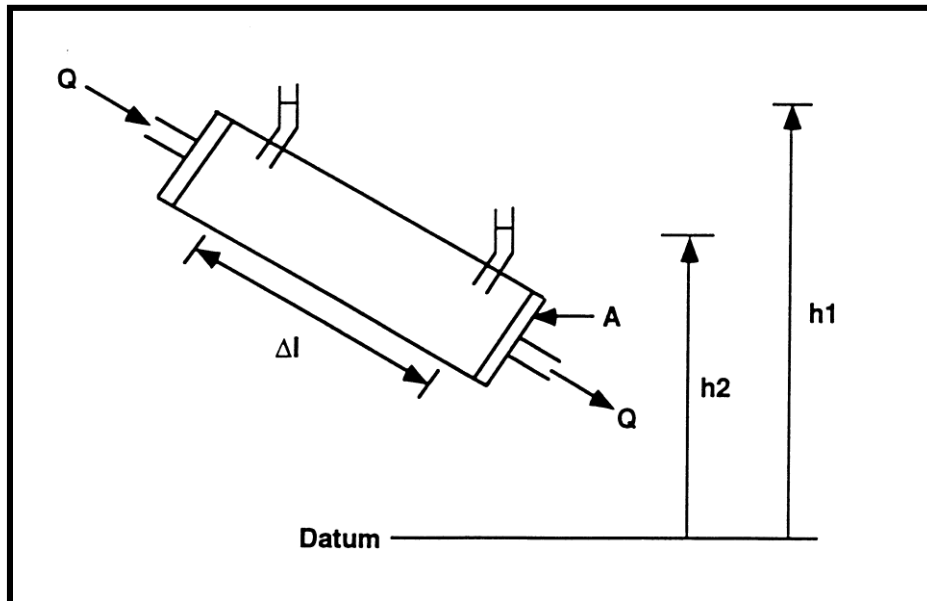


Darcy's Law

In the mid-nineteenth century, Henry Darcy, made the first systematic study of the movement of water through a porous medium. Darcy established empirically that the discharge is proportional to the difference in height of the water and inversely proportional to the flow length.



Darcy's Experiment (1856)

Darcy's Experiments

- Discharge is
 - Proportional to
 - Area
 - Head difference
 - Inversely proportional to
 - Length

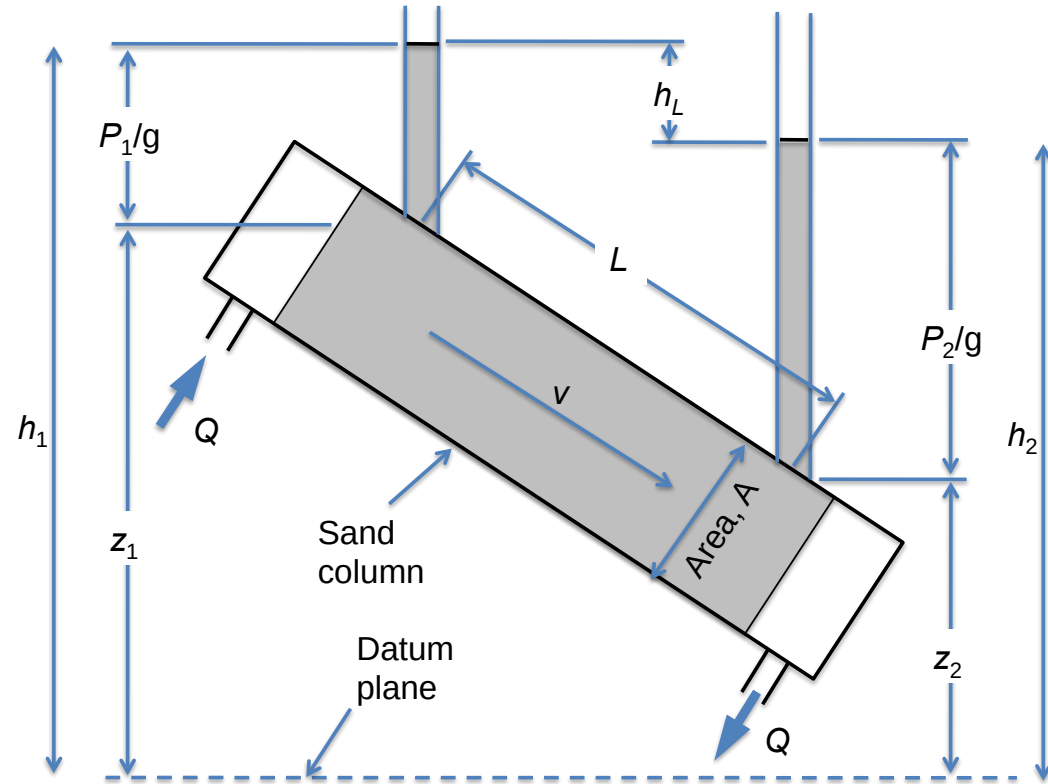
$$Q \propto A \frac{h_1 - h_2}{L}$$

Coefficient of proportionality is
 K = hydraulic conductivity

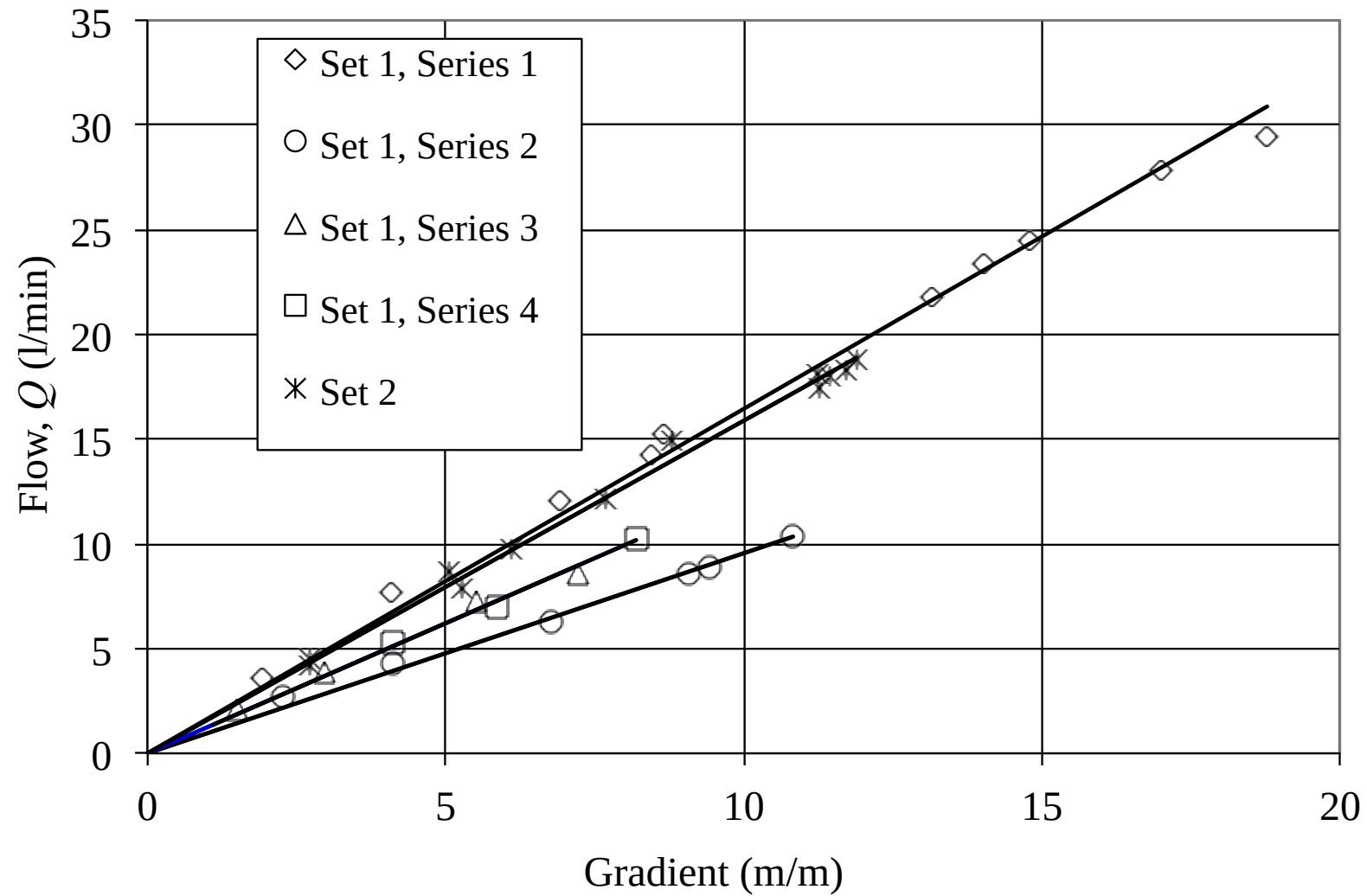
$$Q = -KA \frac{h_2 - h_1}{L}$$

$$Q = -KA \frac{Dh}{L}$$

$$Q = VA \text{ (} A = \text{total area)}$$



Darcy's Data



$$q = \frac{Q}{A} = -K \frac{dh}{dl}$$

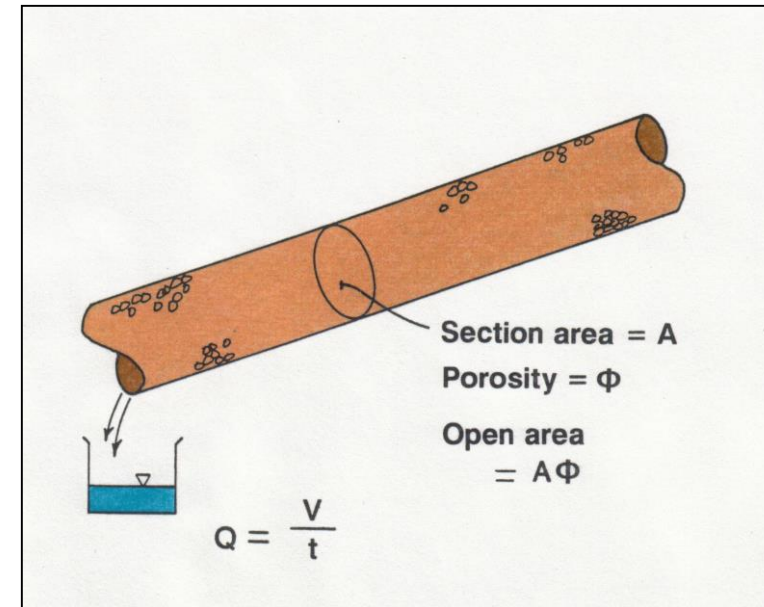
Groundwater Velocity

- q - Specific discharge

Discharge from a unit cross-section area of aquifer formation normal to the direction of flow.

- v - Average velocity

Average velocity of fluid flowing per unit cross-sectional area where flow is ONLY in pores.



$$q = \frac{Q}{A} \quad v = \frac{q}{\phi} = \frac{Q}{A\phi}$$

- From the Continuity Eqn:

$$Q = A v_D = A_V V_s \dots\dots\dots(1)$$

Q = flow rate

A = total cross-sectional area of
material

A_V = area of voids

V_s = seepage velocity

V_D = Darcy velocity

Darcy & Seepage Velocity

- Darcy velocity is a fictitious velocity since it assumes that flow occurs across the entire cross-section of the soil sample.
- Flow actually takes place only through interconnected pore channels.

From eq. (1) $V_S = V_D (A/A_V)$

Multiplying both sides by the length of the medium (L)

$$V_S = V_D (AL / A_V L) = V_D (V_T / V_V)$$

V_T = total volume

V_V = void volume

By Definition, $V_V / V_T = n$, the soil porosity

Thus, $V_S = V_D / n$

Hydraulic Conductivity

Hydraulic conductivity also referred as coefficient of permeability (in older publication), has dimensions of $[L/T^1]$ and is a measure of the ease of movement of water through a porous material. Thus it is not function of physical properties of the porous material.

- Has dimensions of velocity $[L/T^{-1}]$
- A combined property of the medium and the fluid
- Ease with which fluid moves through the medium

$$K = k \frac{\gamma}{\mu}$$

k = cd^2 intrinsic permeability } Porous medium property

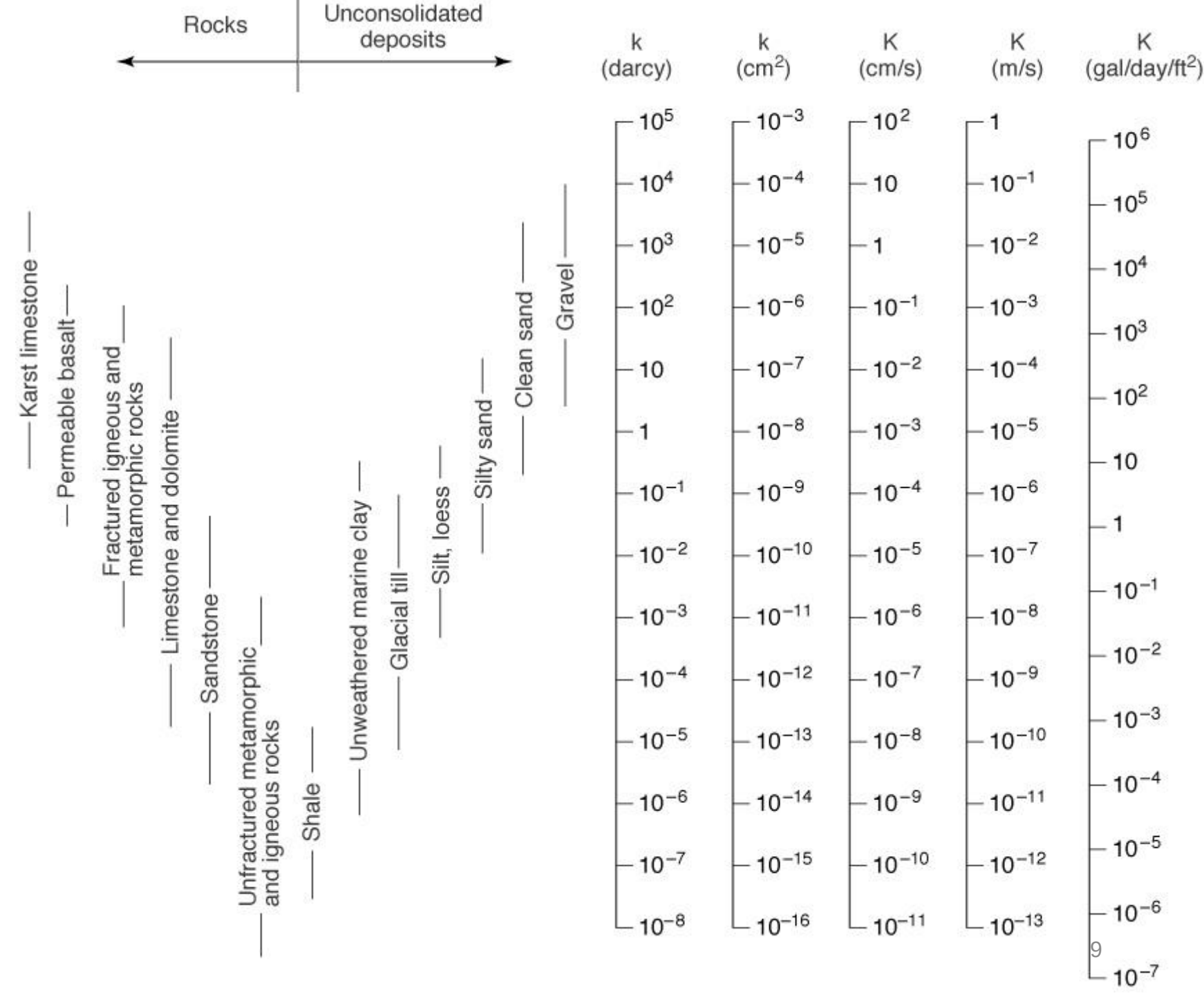
ρ = density

μ = dynamic viscosity

γ = specific weight

} Fluid properties

Hydraulic Conductivity



Validity of Darcy's Law

- We ignored kinetic energy (low velocity)
- We assumed laminar flow
- We can calculate a Reynolds Number for the flow

$$N_R = \frac{r q d_{10}}{m}$$

q = Specific discharge

d_{10} = effective grain size diameter

- Darcy's Law is valid for $N_R < 1$ (maybe up to 10)

LIMITATION OF DARCY'S LAW

We ignored kinetic energy (low velocity)

We assumed laminar flow

Darcy's law is valid for laminar flow, *i.e.*, the Reynolds number (*Re*) varies from 1 to 10, though most commonly it is less than 1

$$Re = \rho V d \div \mu \leq 1$$

Where

V= Velocity of flow through Aquifer

ρ = mass density of water

μ = dynamic viscosity of water

d = mean grain size of the aquifer soil

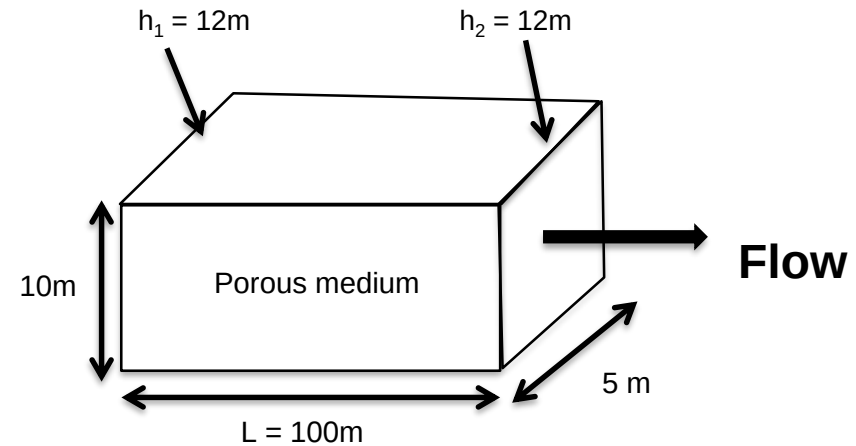
In aquifers containing large diameter solution openings, coarse gravels, rockfills and also in the immediate vicinity of a gravel packed well, flow is no longer laminar due to high gradients and exhibit non linear relationship between the velocity and hydraulic gradient. For example, in a gravel-packed well (mean size of gravel ~ 5 mm) $Re \sim 45$ and the flow would be transitional at a distance of about 5 to 10 times the well radius.

Example Based on Darcy's Law

$$K = 1 \times 10^{-5} \text{ m/s}$$

$$\phi = 0.3$$

Find q , Q , and v



$$dh = (h_2 - h_1) = (10 \text{ m} - 12 \text{ m}) = -2 \text{ m}$$

$$i = dh/dx = (-2 \text{ m})/100 \text{ m} = -0.02 \text{ m/m}$$

$$q = -Ki = -(1 \times 10^{-5} \text{ m/s}) \times (-0.02 \text{ m/m}) = 2 \times 10^{-7} \text{ m/s}$$

$$Q = qA = (2 \times 10^{-7} \text{ m/s}) \times 50 \text{ m}^2 = 1 \times 10^{-5} \text{ m}^3/\text{s}$$

$$v = q/\phi = 2 \times 10^{-7} \text{ m/s} / 0.3 = 6.6 \times 10^{-7} \text{ m/s}$$

Example of Darcy's Law

- A confined aquifer has a source of recharge.
- K for the aquifer is 50 m/day, and n is 0.2.
- The piezometric head in two wells 1000 m apart is 55 m and 50 m respectively, from a common datum.
- The average thickness of the aquifer is 30 m, and the average width of aquifer is 5 km.

Compute:

a) the rate of flow through the aquifer

(b) the average time of travel from the head of the aquifer to a point 4 km downstream

**assume no dispersion or diffusion*

Solution:

- Cross-Sectional area = $30(5)(1000) = 15 \times 10^4 \text{ m}^2$
- Hydraulic gradient = $(55-50)/1000 = 5 \times 10^{-3}$
- Rate of Flow for $K = 50 \text{ m/day}$
-
- $Q = (50 \text{ m/day}) (75 \times 10^1 \text{ m}^2) = 37,500 \text{ m}^3/\text{day}$
- Darcy Velocity:
- $V = Q/A = (37,500 \text{ m}^3/\text{day}) / (15 \times 10^4 \text{ m}^2) = 0.25 \text{ m/day}$

